

Post Lab (Open Ended Practical)

Aim : Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

INPUT :

```
GNU nano 8.7 fault.c
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, f, i, j, k, flag, pageFaults = 0;
    int index = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    // Initialize frames to -1 (empty)
    for(i = 0; i < f; i++) {
        frames[i] = -1;
    }

    printf("\nPage\tFrames\n");
    for(i = 0; i < n; i++) {
        flag = 0;

        // Check if page is already in frame (Hit)
        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
                break;
            }
        }

        // If page not found (Page Fault)
        if(flag == 0) {
            frames[index] = pages[i];
            index = (index + 1) % f; // FIFO replacement
            pageFaults++;
        }

        // Display frames
        printf("%d\t", pages[i]);
        for(k = 0; k < f; k++) {
            if(frames[k] != -1)
                printf("%d ", frames[k]);
            else
                printf("- ");
        }
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", pageFaults);
    return 0;
}
```

OUTPUT :

```
HP@Hiteshri MSYS ~
$ ./virtual
Enter number of pages: 8
Enter page reference string:
1 2 3 4 1 2 5 1
Enter number of frames: 3

Page    Frames
1       1 - -
2       1 2 -
3       1 2 3
4       4 2 3
1       4 1 3
2       4 1 2
5       5 1 2
1       5 1 2

Total Page Faults = 7
```